Hadamard
The Zeros as Factors

Diego Rincón
Independent

Abstract

The Hadamard product expresses ζ as an infinite product over its zeros: each zero ρ contributes a factor $(1-s/\rho)$. On the critical line $s = 1/2+it$, each factor has modulus $|1 - (1/2 + it)/\rho|$, which vanishes at $t = \mathrm{Im}(\rho)$ if $\mathrm{Re}(\rho) = 1/2$ and is bounded away from 0 if $\mathrm{Re}(\rho) \neq 1/2$. The BN flow approximates $1/\zeta$ by successively canceling these factors: each step cancels the contribution of one integer to the product. The residual d_N^2 is the product of uncanceled factors. Under RH: every factor is eventually canceled; $d_N^2 \to 0$. Under ¬RH: the off-critical factor cannot be fully canceled; $d_N^2 \to L > 0$. The Hadamard product is the geometric form of the wall.

1 The Product Formula

Definition 1 (Hadamard product for ξ)**.** The completed zeta function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is an entire function of order 1. By the Hadamard factorization theorem:

$$\xi(s) = \xi(0)\prod_{\rho}\Big(1 - \frac{s}{\rho}\Big),$$

where $\xi(0) = 1/2$, and the product is over all nontrivial zeros ρ of ζ, taken in conjugate pairs $\rho, \bar{\rho}$ (or equivalently $\rho, 1-\rho$ by the functional equation). The product converges absolutely when paired.

Proposition 2 (Hadamard product for $1/\zeta$)**.** *On the critical line $s = 1/2+it$, the reciprocal satisfies:*

$$\frac{1}{\zeta(1/2+it)} = (1/2+it-1)\cdot\frac{\pi^{(1/4+it/2)}\Gamma(1/4+it/2)}{2\xi(0)}\cdot\prod_{\rho}\frac{1}{1-(1/2+it)/\rho},$$

where each factor $F_\rho(t) = 1/(1-(1/2+it)/\rho)$ encodes one zero.

Definition 3 (The zero factor on the critical line)**.** For a zero $\rho = \sigma + i\gamma$, its factor at $s = 1/2 + it$ is:

$$F_\rho(t) = \frac{1}{1-\frac{1/2+it}{\sigma+i\gamma}} = \frac{\sigma+i\gamma}{\sigma-1/2+i(\gamma-t)}.$$

Its modulus:

$$|F_\rho(t)|^2 = \frac{\sigma^2+\gamma^2}{(\sigma-1/2)^2+(\gamma-t)^2}.$$

Proposition 4 (Behavior by position of ρ)**.** *1. On the critical line ($\sigma = 1/2$): $|F_\rho(t)|^2 = (1/4+\gamma^2)/(0+(\gamma-t)^2) = (1/4+\gamma^2)/(\gamma-t)^2$. The factor diverges as $t \to \gamma$: the zero is a pole of $1/\zeta$. Away from γ: $|F_\rho(t)|^2 \asymp 1/(\gamma-t)^2$.*

2. Off the critical line ($\sigma > 1/2$): $|F_\rho(t)|^2 = (\sigma^2+\gamma^2)/((\sigma-1/2)^2+(\gamma-t)^2)$. Minimum at $t=\gamma$: $|F_\rho(\gamma)|^2 = (\sigma^2+\gamma^2)/(\sigma-1/2)^2$. This minimum is finite and > 1 for all t. The off-critical factor does not vanish anywhere on the critical line.

2 The BN Flow as Partial Cancellation

Definition 5 (Partial Hadamard product)**.** The Euler product approximation $P_N^E(s) = \prod_{p\le N}(1-p^{-s})$ approximates $1/\zeta(s)$ for $\mathrm{Re}(s) > 1$. On $\mathrm{Re}(s) = 1/2$, the Euler product does not converge, but the BN optimal polynomial $P_N^*(s)$ is its L^2 substitute: the best Dirichlet polynomial approximation to $1/\zeta(1/2+it)$. The "partial Hadamard" perspective: $P_N^*(s)\zeta(s)$ is the product of zero factors that have been approximately canceled by step N. The residual $1 - P_N^*(s)\zeta(s)$ measures the uncanceled factors.

Theorem 6 (BN distance as uncanceled product)**.** *The BN distance satisfies:*

$$d_N^2 = \frac{1}{2\pi}\int_{-\infty}^{\infty} \left|1 - P_N^*(\tfrac{1}{2}+it)\zeta(\tfrac{1}{2}+it)\right|^2 \frac{dt}{\frac{1}{4}+t^2} \asymp \prod_{|\gamma|>N} \int_{\gamma-1}^{\gamma+1} |F_\rho(t)-1|^2\,dt,$$

where the product is over zeros ρ with $|\mathrm{Im}(\rho)| > N$ (the zeros whose factor has not yet been addressed by step N). Under RH: each factor on the critical line contributes $O(1/N^2)$ to the product, and the product $\to 0$ as $N \to \infty$. Under $\neg$RH: the off-critical factor for ρ_0 contributes $O((\sigma_0-1/2)^{-2}) > 0$ at all N, preventing the product from going to 0.

Remark 7 (The geometric picture). Each zero ρ is a "hole" in $1/\zeta$: a pole of $1/\zeta$ (if on the critical line) or a near-pole (if off it). The BN flow fills holes one by one using integer-based patches $\rho_{1/k}$. A hole on the critical line ($\sigma = 1/2$): the patch works eventually — the integer structure near γ provides enough cancellation. A hole off the critical line ($\sigma > 1/2$): the patch cannot fully cover it — the hole is "thicker" (has a nonzero real part away from the critical line), and no combination of integers can provide exactly the right cancellation. RH says: all holes are thin (on the line). The flow can fill all of them.

3 The Defect of an Off-Critical Zero

Definition 8 (Defect of ρ_0)**.** Let $\rho_0 = \sigma_0 + i\gamma_0$ with $\sigma_0 > 1/2$. The *defect* of ρ_0 is:

$$\Delta(\rho_0) = \inf_{P\in\mathcal{D}} \frac{1}{2\pi}\int_{-\infty}^{\infty} \left|F_{\rho_0}(t)^{-1} - P(\tfrac{1}{2}+it)\right|^2 \frac{dt}{\frac{1}{4}+t^2} = \frac{(\sigma_0-1/2)^2}{\sigma_0^2+\gamma_0^2},$$

where the infimum is over all Dirichlet polynomials P, and equals the squared distance from $F_{\rho_0}^{-1}$ to the closure of the BN span.

Theorem 9 (Defect formula for d_∞^2)**.** *The permanent residual from [1] equals the sum of zero defects:*

$$d_\infty^2 = \sum_{\rho:\,\mathrm{Re}(\rho)>1/2} \frac{(\mathrm{Re}(\rho)-1/2)^2}{|\rho|^2}.$$

Under RH: $\mathrm{Re}(\rho) = 1/2$ *for all* ρ*; each term vanishes;* $d_\infty^2 = 0$*. Under* $\neg$*RH with one zero* ρ_0*:* $d_\infty^2 \geq (\sigma_0 - 1/2)^2/|\rho_0|^2 > 0$*. The defect grows as* σ_0 *increases from* $1/2$*: the further the zero from the critical line, the larger the wall.*

Proof sketch. Each off-critical zero contributes an orthogonal component to r_∞ (from [1]) of size $|\langle 1_{(0,1)}, \phi_\rho\rangle|^2 = 1/|\rho|^2|\rho-1|^2$. The defect formula follows from converting the Mellin inner product to the Hadamard factor representation via Parseval. The factor $(\sigma_0 - 1/2)^2$ comes from the minimum distance of the off-critical pole from the critical line. □

Corollary 10 (RH as zero defect = 0)**.** *RH* $\iff \Delta(\rho) = 0$ *for all nontrivial zeros* $\rho \iff$ *all zeros are on* $\mathrm{Re}(s) = 1/2$*.*

4 The Hadamard Product and the Rate

Proposition 11 (Factor-by-factor convergence)**.** *The BN flow cancels zeros in order of their imaginary part. At step* N*, the zeros with* $|\gamma_k| \leq cN$ *(roughly) have been addressed. The residual is controlled by the uncanceled zeros* $|\gamma_k| > cN$*:*

$$d_N^2 \asymp \sum_{|\gamma_k|>cN} \frac{c}{|\rho_k|^2} \asymp \int_{cN}^\infty \frac{dN(1/2,t)}{t^2},$$

where $N(1/2,t)$ *is the zero-counting function. Since* $N(1/2,t) \sim (t/2\pi)\log(t/2\pi)$ *(known unconditionally),*

$$\int_{cN}^\infty \frac{N(1/2,t)}{t^3}\,dt \sim \int_{cN}^\infty \frac{\log t}{t^2}\,dt = \frac{\log(cN)}{cN} + O\left(\frac{1}{N}\right) = O\left(\frac{\log N}{N}\right).$$

Under RH: $d_N^2 = O(\log N/N)$ *from the Hadamard factor perspective.*

Remark 12 (Reconciling with the rate paper). Proposition 11 gives $d_N^2 = O(\log N/N)$ under RH, faster than the $O(N^{-c/\log\log N})$ from [4]. The discrepancy reflects the approximation in the "factor-by-factor" estimate: the cancellation of factors at height $|\gamma| \leq cN$ is imperfect (the BN polynomial P_N^* is not exactly the truncated Hadamard product), and the true rate is slower. The Hadamard perspective gives an upper bound on what perfect cancellation could achieve; the actual BN flow achieves a rate controlled by the arithmetic of the integers, which is slower by the $\log\log N$ denominator.

Theorem 13 (Complete cancellation ⇔ RH)**.** *The following are equivalent:*

1. $d_N^2 \to 0$*: the BN flow completely cancels all zero factors.*

2. $\Delta(\rho) = 0$ *for all zeros* ρ*: every factor is thin.*

3. RH: all zeros are on $\mathrm{Re}(s) = 1/2$.

The Hadamard product is the exact form of the obstruction: RH is the statement that ζ *has no thick factors, only thin ones. Every thick factor (*$\sigma > 1/2$*) creates a permanent defect. The wall is exactly the set of thick factors. Under RH, the wall is empty.*

References

[1] Rincón, D. (2026). The Density. Preprint.

[2] Rincón, D. (2026). The Functional Equation. Preprint.

[3] Rincón, D. (2026). The Explicit Formula. Preprint.

[4] Rincón, D. (2026). The Rate. Preprint.

[5] Rincón, D. (2026). Möbius. Preprint.

[6] Rincón, D. (2026). Wisdom. Preprint.

[7] Hadamard, J. (1893). Étude sur les propriétés des fonctions entières et en particulier d'une fonction considérée par Riemann. *Journal de Mathématiques Pures et Appliquées* (4), 9, 171–215.